AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT248
Analyzing Data

الجامعة العربية المفتوحة - مصر
Arab Open University

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 14 / Apr. / 2026

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed:** 1.5 hour

Instructions:

- Total Marks: 30
- This exam consists of 2 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1(10 Marks)

Question (1): 5 Marks, 0.5 for each question

Suppose a discrete random variable X has probability function given by

x	3	5	7	8	9	10	12	16
$P(X = x)$	0.08	0.10	0.16	???	0.20	0.03	0.13	0.05

Select the best answer for each of the following:

1- what is the probability that " $X > 8$?"

A) 0.29 B) 0.35 C) 0.68 D) none of these

2- what is the probability that " $X \leq 8$?"

A) 0.34 B) 0.59 C) 0.69 D) none of these

3- what is the probability that " $4 < X < 11$?"

A) 0.65 B) 0.70 C) 0.58 D) none of these

4- what is the mean value " $\mu_x = ?$ "

A) 8.32 B) 10.32 C) 9.32 D) none of these

5- what is the value " $E(X^2) = ?$ "

A) 87.77 B) 0.7778 C) 78.77 D) none of these

6- what is the variance of the distribution " $\sigma_x^2 = ?$ "

A) 77.78 B) 8.32 C) 8.56 D) none of these

7- what is the standard deviation of the distribution " $\sigma_x = ?$ "

A) 8.56 B) 8.82 C) 8.32 D) none of these

8- what is the value " $E(X^3) = ?$ "

A) 801.78 B) 802.78 C) 803.78 D) none of these

9- Given $Y = 40X + 50$, what is the value " $E(Y) = ?$ "

A) 332.80 B) 382.80 C) 383.80 D) none of these

10- From statement number 9, what is the value " $\sigma(Y) = ?$ "

A) 332.80 B) 13696 C) 117.03 D) none of these

Question (2): 5 Marks, 0.5 for each question

Which of the following statements are true/false?

1. For the Exponential Distribution f(x), $p(x > 5) = 1 - \int_0^5 f(x) \ .dx$
2. In a Binomial Distribution, if the probability of success 'p' and failure 'q' are equal, then $\mathbf{p\ (X = x\ )}$ is given as ${}^{n}\mathbf{C}_{x}\ (0.5)^{n}$

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3. If a continuous random variable X has the density function

$$f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & elsewhere \end{cases}$$

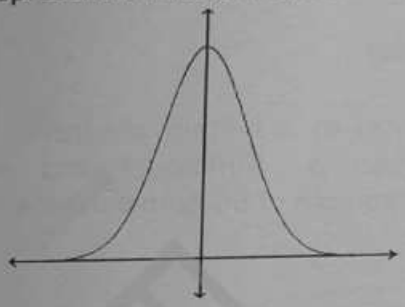
Then the expected value of $e^{2x/3}$ is equal to 7

4. For a laboratory assignment, if the equipment is working, the density function of the observed outcome X is

$$f(x) = \begin{cases} 2(1-x), & 0 < x < 1 \\ 0, & elsewhere \end{cases}$$

The value $E(X^3) = 88.78$

5. Sampling distributions take information from the population to the sample.
6. Is this graph could represent a variable with a normal distribution.



7. The lengths of time that employees of a telecommunications company have worked for the company are normally distributed with a mean of 5.6 years and a standard deviation of 1.9 years. Random samples of size 18 are selected. According to the Central Limit Theorem, the mean of the sampling distribution is 5.6 years and the standard error of the mean is 0.45 years.
8. The mean of the sample means remains constant, and the standard error increases as the sample size decreases.
9. If the population is normally distributed, the sampling distribution of the sample mean will be normal for any sample size.
10. The Central Limit Theorem states that the sampling distribution of the sample mean approaches normality as the sample size increases.

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Part 2: (20 Marks)

Question 3: (5 Marks)

The time between successive breakdowns of a certain machine follows an exponential distribution with a mean of 17 days.

a) (1 mark)
Find the value of the parameter λ of the distribution.

b) (2 marks)
Write down the probability density function (pdf) of the time T between breakdowns.

c) (2 marks)
Find the probability that the machine breaks down within the first 15 days.
Give your answer to **4 decimal places**.

Question 4: (5 Marks)

The lifetime (in hours) of a certain electronic component is normally distributed with mean $\mu = 500$ hours and standard deviation $\sigma = 80$ hours. A random sample of **64 components** is selected.

(a) *(1 mark)*
State the mean and the standard deviation of the sampling distribution of the sample mean $\bar{X}$.

(b) (1 marks)
Find the probability that the lifetime exceeds 520 hours.

(c) (1 marks)
Find the probability that the sample mean lifetime exceeds 520 hours.

(d) (2 marks)
Find the value k such that $P(\bar{X} < k) = 0.90$

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Question 5: (10 Marks)

A population consists of four numbers: 3, 4, 5 and 6. Consider all possible distinct samples of size 2 with replacement. Find

a. Population mean **(1 marks)**
b. Population Standard deviation **(1 marks)**
c. Sampling distribution of mean **(4marks)**
d. Mean of Sampling distribution of mean **(1 marks)**
e. Standard deviation of Sampling distribution of mean **(1 marks)**
f. Verify (c) and (e) directly from (a) and (b) by using appropriate formulas. **(2 marks)**

End of Question

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Appendix A: Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

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